EE511 Protection of Power Systems II

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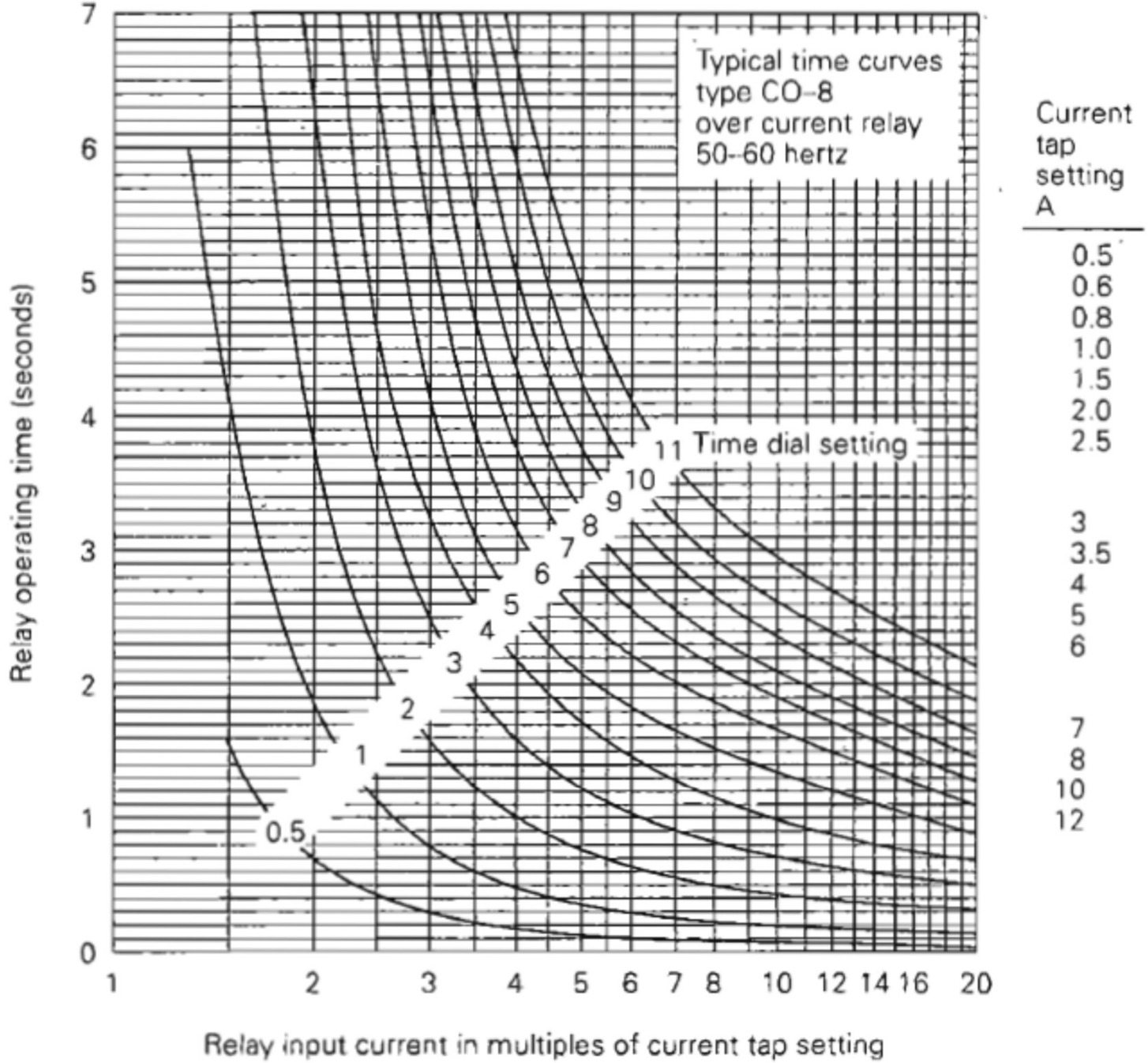
slides 2-11 are from Chapter 10, pages 533-541, of "Power System Analysis and Design", Fifth Edition, 2012, J. D. Glover, M. S. Saema, T. J. Overbye
slides 2-11 are from "Protective relaying theory and applications" by Walter A. Elmore

Standard CT ratios

Current Ratios						
50:5	100:5	150:5	200:5	250:5	300:5	400:5
450:5	500:5	600:5	800:5	900:5	1000:5	1200:5
1500:5	1600:5	2000:5	2400:5	2500:5	3000:5	3200:5
4000:5	5000:5	6000:5				

OVERCURRENT RELAYS



Operating time for a CO-8 time-delay overcurrent relay

The CO-8 relay with a current tap setting of 6 amperes and a time-dial setting of 1 is used with the 100:5 CT. Determine the relay operating time for each case.

$$I = 5\text{ A} \quad I = 8\text{ A} \quad I = 15\text{ A}$$

SOLUTION

a. From

$$I = 5\text{ A} \qquad \frac{I}{I_p} = \frac{5}{6} = 0.83$$

The relay does not operate. It remains in the blocking position

b.

$$I = 8\text{ A} \qquad \frac{I}{I_p} = \frac{8}{6} = 1.33$$

Using curve 1, $t_{operating} = 6$ seconds.

c.

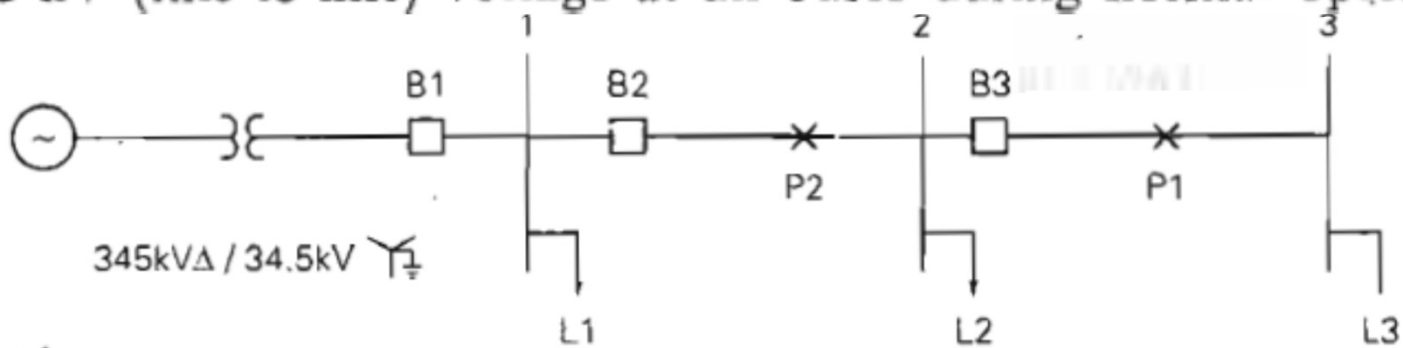
$$I = 15\text{ A} \qquad \frac{I}{I_p} = \frac{15}{6} = 2.5$$

From curve 1, $t_{operating} = 1.2$ seconds.

RADIAL SYSTEM PROTECTION

Select current tap settings (TSs) and time-dial settings (TDSs) to protect the system from faults. Assume three CO-8 relays for each breaker, with a 0.3-second coordination time interval.

Assume a 34.5-kV (line-to-line) voltage at all buses during normal operation.



Maximum loads-

Bus	S MVA	Lagging p.f.
1	11.0	0.95
2	4.0	0.95
3	6.0	0.95

fault currents

Bus	Maximum Fault Current (Bolted Three-Phase) A	Minimum Fault Current (L–G or L–L) A
1	3000	2200
2	2000	1500
3	1000	700

Breaker, CT, and relay data

Breaker	Breaker Operating Time	CT Ratio	Relay
B1	5 cycles	400:5	CO-8
B2	5 cycles	200:5	CO-8
B3	5 cycles	200:5	CO-8

Maximum loads

Bus	S MVA	Lagging p.f.
1	11.0	0.95
2	4.0	0.95
3	6.0	0.95

fault currents

Bus	Maximum Fault Current (Bolted Three-Phase) A
1	3000
2	2000
3	1000

SOLUTION First, select TSs such that the relays do not operate for maximum load currents. Starting at B3, the primary and secondary CT currents for maximum load L3 are

$$I_{L3} = \frac{S_{L3}}{V_3\sqrt{3}} = \frac{6 \times 10^6}{(34.5 \times 10^3)\sqrt{3}} = 100.4 \text{ A}$$

$$I'_{L3} = \frac{100.4}{(200/5)} = 2.51 \text{ A}$$

From Figure 10.12, we select for the B3 relay a 3-A TS, which is the lowest TS above 2.51 A.

Note that $|S_{L2} + S_{L3}| = |S_{L2}| + |S_{L3}|$ because the load power factors are identical. Thus, at B2, the primary and secondary CT currents for maximum load are

$$I_{L2} = \frac{S_{L2} + S_{L3}}{V_2\sqrt{3}} = \frac{(4 + 6) \times 10^6}{(34.5 \times 10^3)\sqrt{3}} = 167.3 \text{ A}$$

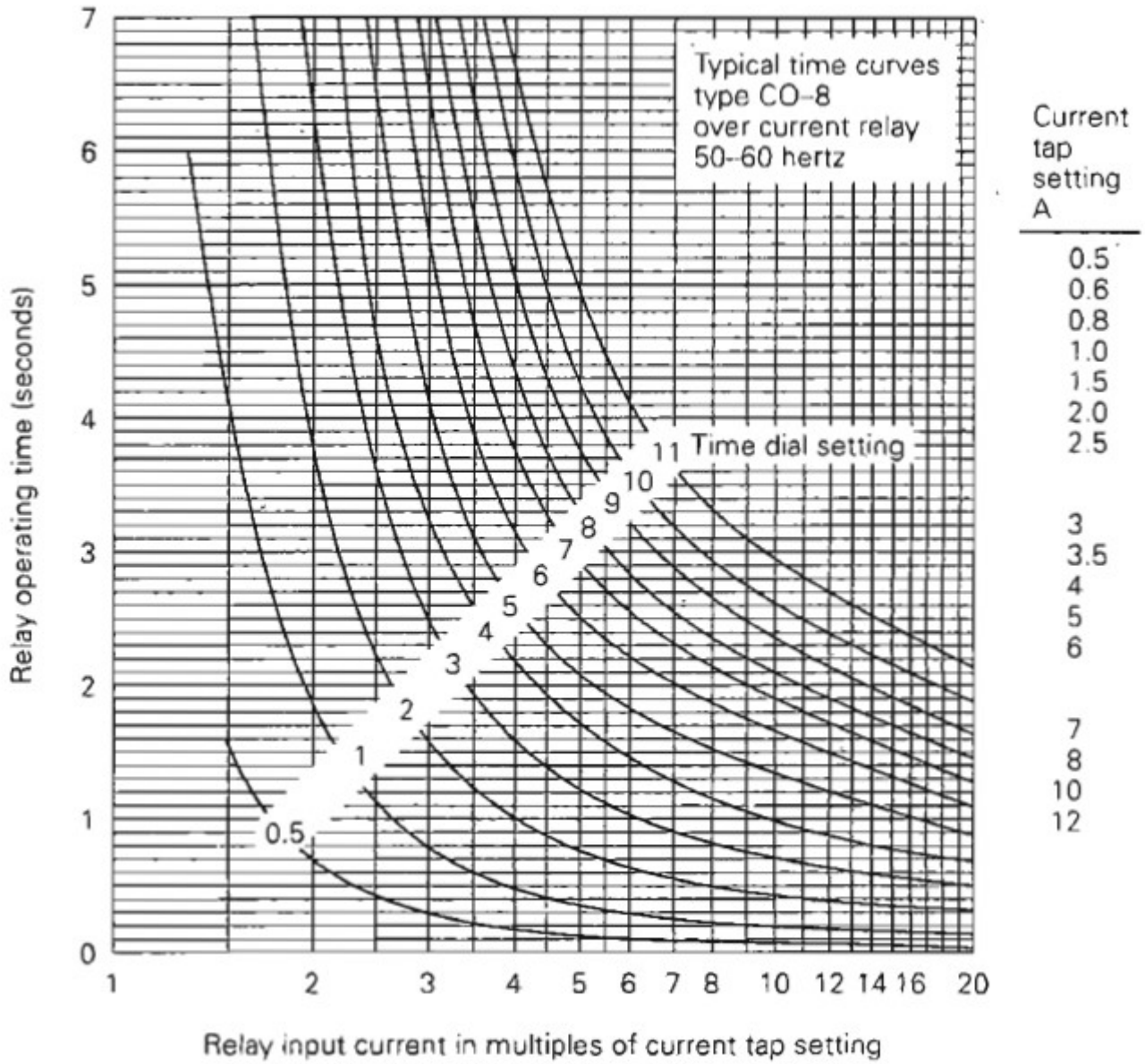
$$I'_{L2} = \frac{167.3}{(200/5)} = 4.18 \text{ A}$$

From Figure 10.12, select for the B2 relay a 5-A TS, the lowest TS above 4.18 A. At B1,

$$I_{L1} = \frac{S_{L1} + S_{L2} + S_{L3}}{V_1\sqrt{3}} = \frac{(11 + 4 + 6) \times 10^6}{(34.5 \times 10^3)\sqrt{3}} = 351.4 \text{ A}$$

$$I'_{L1} = \frac{351.4}{(400/5)} = 4.39 \text{ A}$$

Select a 5-A TS for the B1 relay.


Typical time curves
type CO-8
over current relay
50-60 hertz
Time dial setting
11
10
9
8
7
6
5
4
3
2
1
0.5
Relay operating time (seconds)
0
1
2
3
4
5
6
7
1
2
3
4
5
6
7
8
10
12
14
16
20
Relay input current in multiples of current tap setting
Current tap setting A
0.5
0.6
0.8
1.0
1.5
2.0
2.5
3
3.5
4
5
6
7
8
10
12

Starting at B3, the largest fault current through B3 is 2000 A, which occurs for the three-phase fault at bus 2 (just to the right of B3). Neglecting CT saturation, the fault-to-pickup current ratio at B3 for this fault is

$$\frac{I'_{3Fault}}{TS3} = \frac{2000/(200/5)}{3} = 16.7$$

Since we want to clear faults as rapidly as possible, select a 1/2 TDS for the B3 relay. Then, from the 1/2 TDS curve in Figure 10.12, the relay operating time is T3 = 0.05 seconds. Adding the breaker operating time (5 cycles = 0.083 s), primary protection clears this fault in $T3 + T_{breaker} = 0.05 + 0.083 =$ 0.133 seconds.

For this same fault, the fault-to-pickup current ratio at B2 is

$$\frac{I'_{2Fault}}{TS2} = \frac{2000/(200/5)}{5} = 10.0$$

Adding the B3 relay operating time (T3 = 0.05 s), breaker operating time (0.083 s), and 0.3 s coordination time interval, we want a B2 relay operating time

$$T2 = T3 + T_{breaker} + T_{coordination} = 0.05 + 0.083 + 0.3 \approx 0.43 \text{ s}$$

From Figure 10.12, select TDS2 = 2.

Next select the TDS at B1. The largest fault current through B2 is 3000 A, for a three-phase fault at bus 1 (just to the right of B2). The fault-to-pickup current ratio at B2 for this fault is

$$\frac{I'_{2Fault}}{TS2} = \frac{3000/(200/5)}{5} = 15.0$$

From the 2 TDS curve in Figure 10.12, T2 = 0.38 s. For this same fault,

$$\frac{I'_{1Fault}}{TS1} = \frac{3000/(400/5)}{5} = 7.5$$

$$T1 = T2 + T_{breaker} + T_{coordination} = 0.38 + 0.083 + 0.3 \approx 0.76 \text{ s}$$

From Figure 10.12, select TDS1 = 3. The relay settings are shown

Breaker	Relay	TS	TDS
B1	CO-8	5	3
B2	CO-8	5	2
B3	CO-8	3	1/2

Distribution Feeder Protection

Selecting an Overcurrent Relay Time Curve

Five different curve shapes have been established by the vast number of electromechanical relays that are in service on power systems and that dictate coordination requirements.

1. Definite time, CO-6
2. Moderately inverse, CO-7
3. Inverse, CO-8
4. Very inverse, CO-9
5. Extremely inverse, CO-11

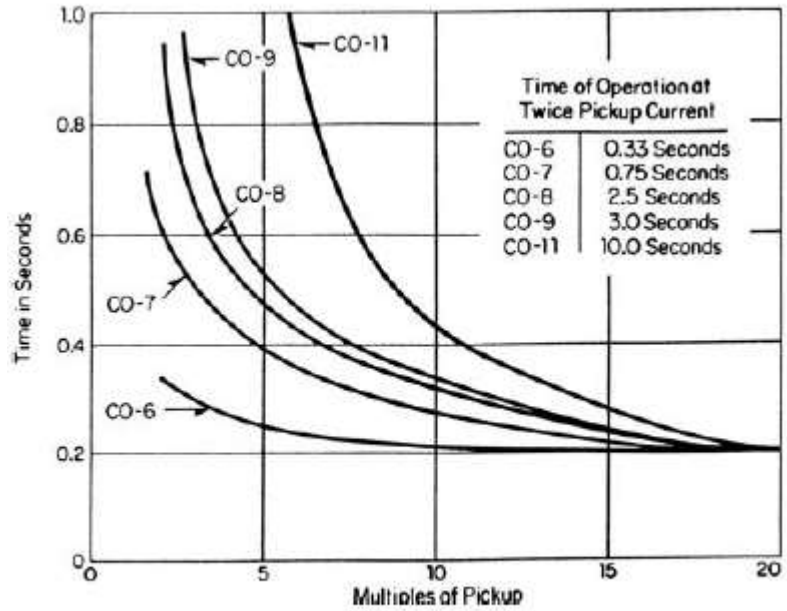


Figure 12-3 Type CO curve shape comparison.

The time lever settings are selected so that all relays operate in 0.2 sec at 20 times the tap setting.

Protective relaying theory and applications, by Walter A. Elmore

Distribution Feeder Protection

1. The flatter curves (CO-6 and CO-7) are more suitable when:

a) There are no coordination requirements with other types of protection devices farther out in the system.
b) The variation in current for faults at the near and far ends of the protected circuit is too small to take advantage of the inverse characteristic.

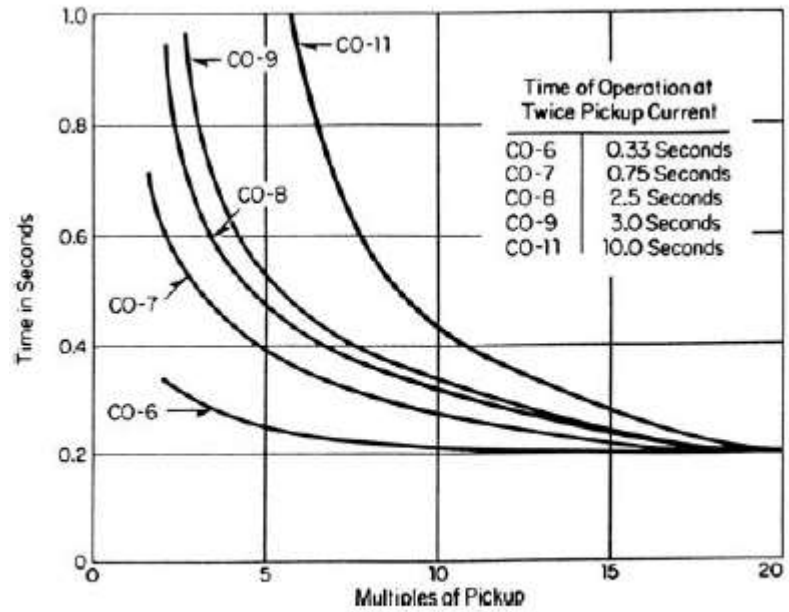


Figure 12-3 Type CO curve shape comparison.

Distribution Feeder Protection

2. The "inverse-time" relay (CO-8) provides faster clearing time than the "more inverse-time" relays for low-current faults. This would be advantageous on long lines where the available fault current is much less at the end of the line than at the local end. It does not provide much margin for cold load pickup.

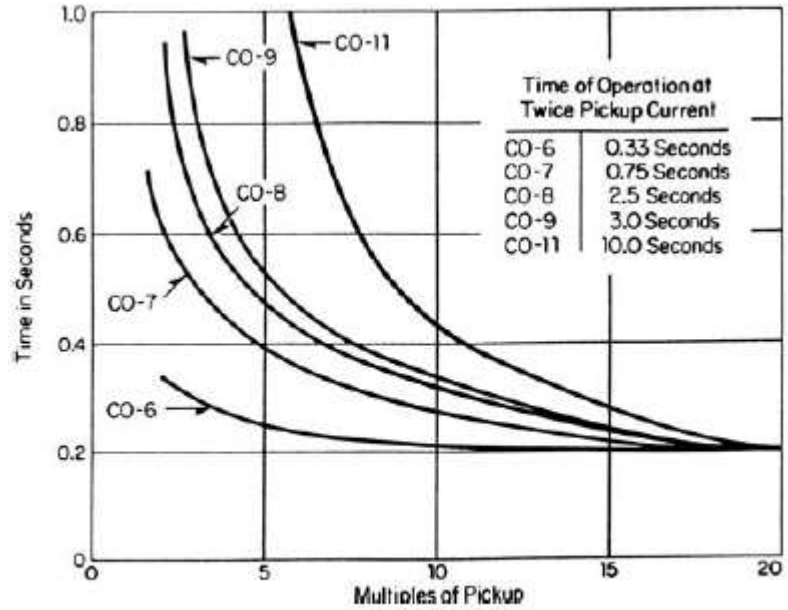


Figure 12-3 Type CO curve shape comparison.

Distribution Feeder Protection

3. The steeper (more inverse) curves (CO-9 and CO-11) are more suitable when

a) Fault currents are significantly different for the close-in and remote faults (for example, when the line impedance is large compared to the source impedance).
b) There is an appreciable current inrush on service restoration (cold load).
c) Coordination with other types of devices with very inverse characteristics, such as fuses and reclosers, is required.

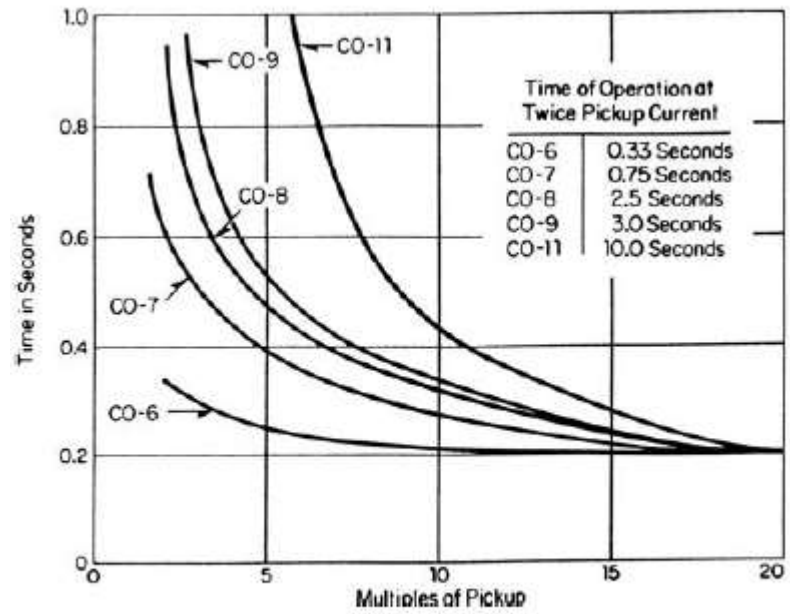


Figure 12-3 Type CO curve shape comparison.

Distribution Feeder Protection

General Comments on Curve Shape Selection

1. Use a CO-6 definite minimum time relay when coordination is not a problem.
2. Use a CO-6 relay for short line application.
3. Use a CO-11, extreme inverse-time relay when fuse are involved.
4. The more inverse shape (CO-8, CO-9, and CO-11) is more suitable in loop systems.
5. Use a comparable shape within a system segment for easier coordination.

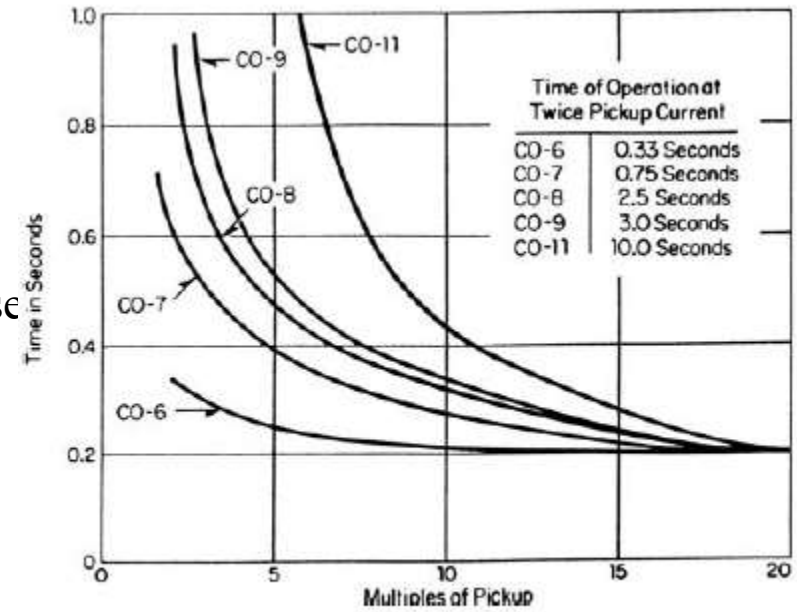


Figure 12-3 Type CO curve shape comparison.

One of the advantages of using a microprocessor-based overcurrent relay is that a different time curve can be selected in the same device without changing the unit.